

Foundation Program in Literacy, Numeracy and Skills

NUMERACY

Teacher's Guide: GRADE 9 –Lesson 6 Area of circle (Les Métiers)

Competency, Element and Performance Criteria:

C2.2 Application of geometry in practical situations and trades.

Element 4: Find area of a circle.

Performance Criteria:

2.2.4A Circles

Level 2: Find area of a circle.

Purpose of activity:

To make learning engaging and relevant by connecting mathematical concepts to real-world contexts.

Learning Outcomes:

By the end of the lesson, students should be able to

- Demonstrate understanding of **area of a circle** ($A = \pi r^2$).
- Apply the formula to solve real-world problems related to various professions (**Les Métiers**).
- Develop an appreciation for how mathematics is used in different careers.

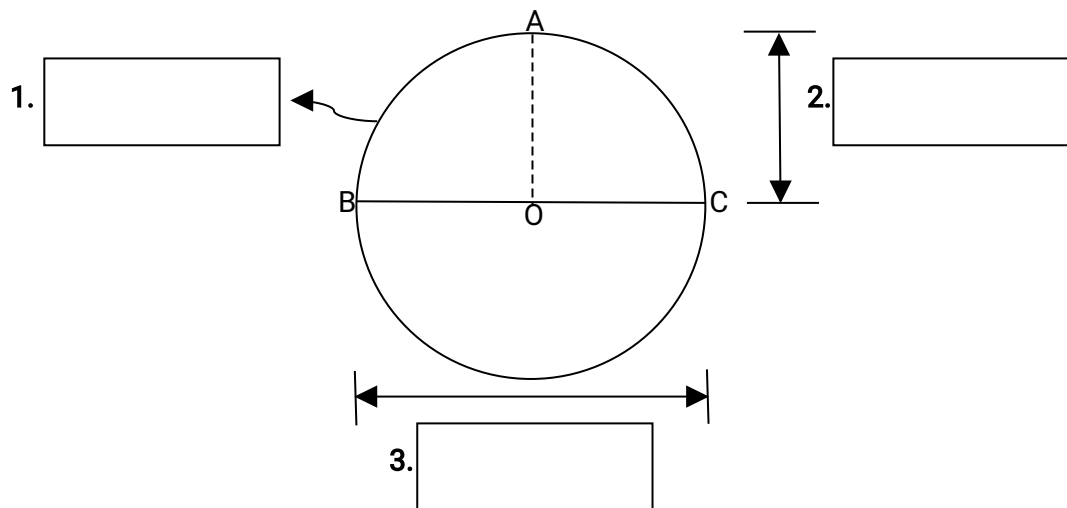
Resources and materials:

- **Visual Aids** – Charts or posters showing the formula: $A = \pi r^2$
- **Manipulatives** – Circular objects like coins, lids, or paper cutouts to help students understand radius and diameter
- **Grid Paper** – Helps students visualize area by counting squares within a drawn circle
- **Compasses & Rulers** – For students to practice drawing circles and measuring radius/diameter.

Teaching trajectories/Implementation guidelines:

Activity 1 (Recall: Circle, radius, diameter and circumference)

Fill in the boxes using the words **radius**, **diameter** and **circumference**.



Activity 2 - Warm- Up: Professions and Circles



Objective: Engage students with real- world examples

- Start by showing pictures of objects related to professions:
 - **Chef** – Pizza, cakes, plates
 - **Mechanic** – Car wheels, gears
 - **Architect** – Round buildings, fountains
 - **Watchmaker** – Clock faces

Discuss: Why do certain professions work with circular shapes?

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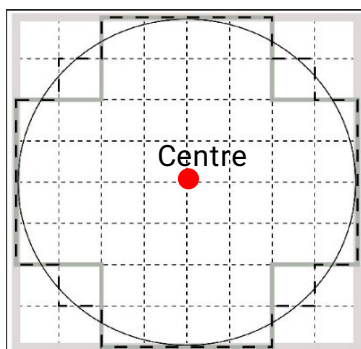
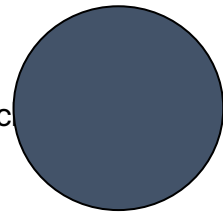
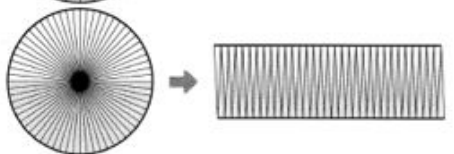
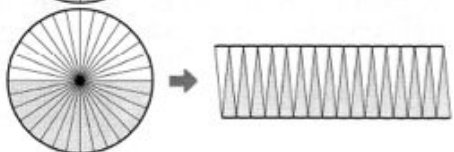
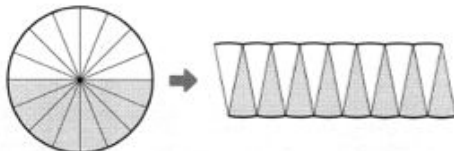
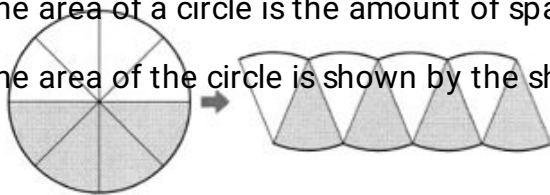
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Activity 3

Gently introduce the meaning of area of a circle

The area of a circle is the amount of space that is inside the circle.

The area of the circle is shown by the shaded region.



Count the number of complete and incomplete squares inside the circle.

Number of complete squares = 33

Number of incomplete squares = 22

Assume that two incomplete squares are equal to one complete square.

Then, 22 incomplete squares = 11 complete squares.

Total number of complete squares = $33 + 11 = 44$

Example: Use of formula $A = \pi r^2$ to find area of a circle

A circular coin has a radius of 4 cm. What is the area of the coin? (Take $\pi = 3.14$)

Steps to Solve:

1. Write the formula:

$$\text{Area} = \pi r^2$$

2. Substitute the radius (r=4):

$$\text{Area} = 3.14 \times (4)^2$$

3. Square the radius:

$$\text{Area} = 3.14 \times 16$$

4. Multiply:

$$\text{Area} = 50.24 \text{ cm}^2$$